# Binary Tree Diameter

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# The problem

- Number of edges on the longest path between any two nodes.

# Intuition

- The longest path through a node joins the heights of its two subtrees.

# Approach

- 1. Post-order returns each subtree's height
- 2. At each node, left height plus right height is a candidate
- 3. Track the maximum

# Java solution

```java
public class BinaryTreeDiameter {
    public static int binaryTreeDiameter(BinaryTree t) {
        int[] best = {0};
        height(t, best);
        return best[0];
    }

    private static int height(BinaryTree t, int[] best) {
        if (t == null) return 0;
        int l = height(t.left, best), r = height(t.right, best);
        best[0] = Math.max(best[0], l + r);
        return 1 + Math.max(l, r);
    }
}
```

# Complexity

- Time: O(n)
- Space: O(h)

# Recap

- The longest path through a node joins the heights of its two subtrees.
- Complexity: O(n)